Yiming Yang

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EDUCATION

Ph.D. in Electrical and Computer Engineering Expected May 2026

Michigan Technological University, Houghton, MI, USA

- Thesis: Robust Autonomous Driving in Winter Weather: Perception, Planning, and Control Under Adverse Conditions

Master in Control Science and Engineering Sep. 2017 - Jul. 2020

Chang'an University, Xi'an, China

- Thesis: Constrained Optimization and Distributed Model Predictive Control-Based Merging Strategies for Adjacent Connected Autonomous Vehicle Platoons

Bacholar in Automation Sep. 2013 - Jul. 2017

Chang'an University, Xi'an, China

- Graduated from the "Excellent Engineer" honors program

RESEARCH INTERESTS

- **Perception system assessment under adverse winter conditions:** This study aims to quantify LiDAR-based 3D object detection performance degradation in snow, using a large-scale winter autonomy dataset to analyze the combined impact of snowfall intensity, range, and scene type.
- **Navigation in snow-covered & white-out conditions:** Robust winter navigation and traversability estimation by detecting the unique thermal properties of wheel tracks in visually degraded, snow-covered environments using RGB-Thermal (RGB-T) sensor fusion in white-out conditions.
- **Autonomous control on snow and ice:** Investigate vehicle performance limits on snow and ice using a neural network dynamics model and a Model Predictive Path Integral (MPPI) controller, validate on a 1:5 scale autonomous testbed to quantify performance degradation in winter.

RESEARCH EXPERIENCE

Proposal: Robust 3D Human Pose Estimation and Tracking in Adverse Winter Conditions through Thermal and VIS Fusion Submitted Sep. 2025

- Role: Lead Author and Researcher.
- Sponsor: Sony Research Award Program.

1. **SAE Autodrive Challenge II** 2022 - present
   - Role: Graduate Teaching Assistant (GTA) of teams: Controls and Mathworks, Safety and Testing.
   - Sponsor: General Motors & SAE International.
   - Contributions: Architected the vehicle control interface with dSPACE MicroAutoBox III; designed and implemented the ROS-dSPACE communication pipeline; developed a safety-critical autonomy state machine with robust failure handling and safe-exit mechanisms; led HIL and VIL testing and subsystem validation on a dyno; authored key safety documentation, e.g., DVP&R.
2. **The Railroad Crossing Vehicle Warning (RCVW) system** 2021 - 2022
   - Role: Graduate Research Assistant.
   - Sponsor: Federal Railroad Administration (FRA).
   - Contribution: Led the system implementation, including the configuration of VBS/RBS/RSU hardware and software, railroad crossing map creation, and field testing; developed a HIL virtual demonstration system to replicate real-world scenarios; created and presented large-scale virtual demonstrations of the RCVW application for Connected Vehicles using PTV VISSIM for the Rail Learning System (RLS) and FRA website.

3. Integration of Unmanned Aerial Systems Data Collection Into Day-to-Day Usage for Transportation Infrastructure – A Phase III Project 2021 - 2022

- Role: Graduate Research Assistant.
- Sponsor: Michigan Department of Transportation (MDOT).
- Contributions: Executed detailed data labeling on the Williamston Road (I-96) traffic corridor dataset to train a machine-learning model for autonomous vehicle counting; subsequently analyzed the traffic data to generate OD insights and validate vehicle counts.

4. Microscopic Traffic Safety Analysis and macro-level transportation analysis 2021 - 2022

- Role: Graduate Research Assistant.
- Affiliation: Dept. of Civil, Environmental, and Geospatial Engineering, Michigan Tech.
- Contributions: Conducted a microscopic safety analysis on the Waymo dataset by visualizing agent dynamics against road structures, extracting lane geometries to identify intersection conflict points, and generating lane polygons for precise map-matching; additionally, performed a macro-level transportation analysis using the Weijo dataset to conduct an OD study, identifying key mobility trends and peak-hour congestion zones.

5. From OSM to VISUM: Automated Infrastructure Generation for Transport Planning. 2021 - 2022

- Role: Graduate Research Assistant.
- Affiliation: Dept. of Civil, Environmental, and Geospatial Engineering, Michigan Tech.
- Contributions: Developed an automated pipeline to convert OpenStreetMap data (nodes, ways, tags) into simulation-ready PTV VISUM networks while ensuring network connectivity, generated Points of Interest and automated zone divisions, and exported the final topology into VISUM-compatible formats (.net/.ver). This automation reduced manual network-building time from days to minutes, enabling rapid scenario testing.

6. From OSM to VISSIM: Automating Large-Scale Traffic Network Modeling 2021 - 2022

- Role: Graduate Research Assistant.
- Affiliation: Dept. of Civil, Environmental, and Geospatial Engineering, Michigan Tech.
- Contributions: Developed an automated pipeline converting OSM data into high-fidelity PTV VISSIM models; interpreted road class, lane counts, and speed limits to model MUTCD-compliant intersections and utilized spline-based connectors for precise vehicle paths. The process generated native (.inpx) files, enabling rapid creation of large-scale, high-resolution traffic models.

7. Research on Construction and Site Testing Technology of Closed Test Environment for Auto-driving Electric Vehicles 2017 - 2020

- Role: Graduate Research Assistant.
- Sponsor: National Key Research and Development Program.
- Contributions: Led hardware and software adaptation to transition the autonomous system from gasoline to EV architecture; developed an single-chip microcomputer based vehicle controller, and engineered and calibrated precise drive-by-wire algorithms for steering, propulsion, and braking.

8. QLTD Intelligent and Connected Expressway Testbase construction 2017 - 2020

- Role: Graduate Research Assistant.
- Sponsor: QiLu Transportation Development Group and Chang'an University.
- Contributions: Implement V2X systems for both control - an adaptive IDM and a cellular-based remote control system; and safety - developed a V2X-based forward collision warning system. All functionalities were validated through real-world highway platoon tests using V2V (DSRC/C-V2X).

9. Key Technologies for Driverless Commercial Vehicle Control Systems 2017 - 2020

- Role: Graduate Research Assistant.
- Sponsor: Key Projects of Key Research and Development Programs in Shaanxi Province.
- Contributions: Designed and implemented a by-wire motion-control system (throttle, brake, steering) using incremental PID algorithms; deployed trajectory tracking algorithms including Pure Pursuit and MPC;

additionally, engineered a safety-critical remote controller with override/E-stop functionality and power supply system for all onboard electronics.

PUBLICATIONS

1. Tracking the Invisible: Self-Supervised Wheel Track Detection with Thermal-RGB Fusion May 2025

Yiming Yang, Jeremy P. Bos - Under review at IEEE Transactions on Intelligent Vehicles (T-IV), 2025

2. Aggressive Autonomous Control on Snow and Ice April 2025

Yiming Yang, Jeremy P. Bos

SAE International Journal of Advances and Current Practices in Mobility (10.4271/2025-01-8040)

3. Constrained optimization and distributed model predictive control-based merging strategies for adjacent connected autonomous vehicle platoons Nov. 2019

Haigen Min, Yiming Yang, Yukun Fang, Pengpeng Sun, Xiangmo Zhao, 10.1109/ACCESS.2019.2952049

TALKS & PRESENTATIONS

1. Aggressive Autonomous Control on Snow and Ice April 2025

Conference proceedings talk, World Congress Experience (WCX), 2025, Detroit, MI, USA

2. Deep Deterministic Policy Gradient based Cooperative Platoon Logitudinal Control Strategy Jan. 2021

Conference presentation: Transportation Research Board (TRB), 2021, Virtual

3. Research on Automatic Emergency Line Change Control Strategy Based on MPC Jun. 2019

Conference presentation, 2019, Beijing, China

PATENTS

1. Autonomous vehicle controllers (NO: ZL 2019 2 0786357.5)

2. Autonomous vehicle platoon controller (NO: ZL 2019 2 0784716.3)

3. Unmanned vehicle power system and unmanned vehicle (NO: ZL 2019 2 1037928)

SELECTED HONORS & AWARDS

1. Excellent graduate of Chang'an University June 2020
2. First-class Academic Scholarship 2017-2018 Nov. 2018
3. 2017-2018 University-level Outstanding Graduate Student Nov. 2018
4. Best Technology Award of 2018 i-VISTA Auto Driving Challenge Aug. 2018
5. First Prize of 2018 i-VISTA Auto Driving Challenge - Urban Traffic Scene Challenge Aug. 2018
6. Second Prize of 2018 China Robot Competition - General Service Robot Project Aug. 2018
7. First-class academic scholarship 2016-2017 Nov. 2017
8. University-level first prize: Electric vehicle charging pile and operation management platform, Team award Jan. 2017

SKILLS

Technical Skills: Computational: Python, C/C++, MATLAB/Simulink for algorithm development; **Control Systems:** Real-time control design and implementation (dSPACE); **Robotics:** System integration with ROS; **Hardware:** CAN bus analysis, CAD (SOLIDWORKS), and PCB design (Altium Designer).